The Exact Bochnak Complexification Constant of the Real ℓ_1 Plane

Full answer to the exact-value question in arXiv:2311.03531

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Status: candidate full solution. The argument gives matching upper and lower bounds. The lower bound uses the classical weighted Siciak–Zakharyuta theorem after an explicit verification that the relevant harmonic function glues subharmonically across the unit circle. Independent expert review is recommended.

1 Source question

Rodríguez defines the Bochnak complexification constants

$$c_{\mathrm{b}}(k, \ell_1^2(\mathbb{R})) = \sup\{\|\widetilde{P}\| : P \in \mathcal{P}({}^k\ell_1^2(\mathbb{R})),\ \|P\| \le 1\}$$

and

$$c_{\mathrm{b}}(\ell_1^2(\mathbb{R})) = \limsup_{k\to\infty} c_{\mathrm{b}}(k, \ell_1^2(\mathbb{R}))^{1/k}.$$

The source says that its exact value is unknown and records only $2^{1/4} \le c_{\mathrm{b}}(\ell_1^2(\mathbb{R})) \le \sqrt{2}$ [1, p. 3].

2 Result

Let

$$G = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} = 0.915965594177219\ldots$$

be Catalan's constant.

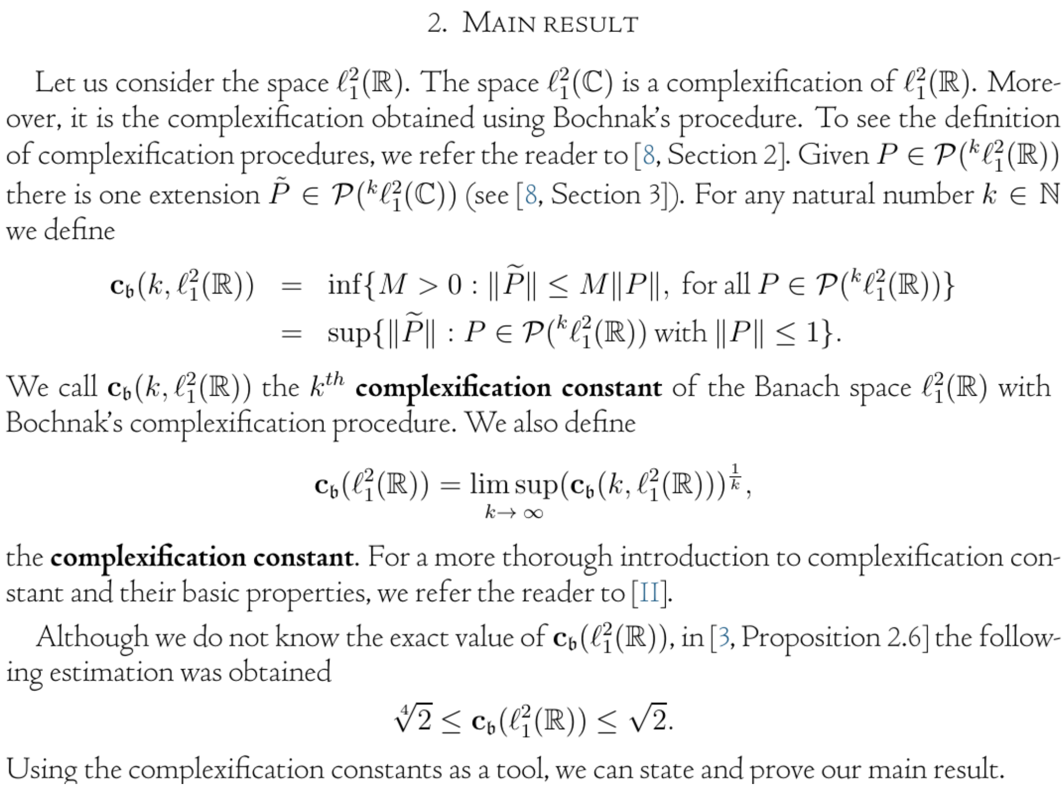
Theorem 1. *The exact Bochnak complexification constant of the real ℓ_1 plane is*

$$\boxed{c_{\mathrm{b}}(\ell_1^2(\mathbb{R})) = \frac{e^{2G/\pi}}{\sqrt{2}}} = 1.2668686397429209\ldots.$$

In fact, for every $k \ge 1$,

$$c_{\mathrm{b}}(k, \ell_1^2(\mathbb{R})) \le \left(\frac{e^{2G/\pi}}{\sqrt{2}}\right)^k,$$

and a sequence of real homogeneous polynomials asymptotically attains this bound after taking kth roots.



2. MAIN RESULT

Let us consider the space $\ell_1^2(\mathbb{R})$. The space $\ell_1^2(\mathbb{C})$ is a complexification of $\ell_1^2(\mathbb{R})$. Moreover, it is the complexification obtained using Bochnak's procedure. To see the definition of complexification procedures, we refer the reader to [8, Section 2]. Given $P \in \mathcal{P}(^k\ell_1^2(\mathbb{R}))$ there is one extension $\tilde{P} \in \mathcal{P}(^k\ell_1^2(\mathbb{C}))$ (see [8, Section 3]). For any natural number $k \in \mathbb{N}$ we define

$$\begin{aligned} \mathbf{c}_{\mathfrak{b}}(k, \ell_1^2(\mathbb{R})) &= \inf\{M > 0 : \|\widetilde{P}\| \leq M\|P\|, \text{ for all } P \in \mathcal{P}(^k\ell_1^2(\mathbb{R}))\} \\ &= \sup\{\|\widetilde{P}\| : P \in \mathcal{P}(^k\ell_1^2(\mathbb{R})) \text{ with } \|P\| \leq 1\}. \end{aligned}$$

We call $\mathbf{c}_{\mathfrak{b}}(k, \ell_1^2(\mathbb{R}))$ the k^{th} **complexification constant** of the Banach space $\ell_1^2(\mathbb{R})$ with Bochnak's complexification procedure. We also define

$$\mathbf{c}_{\mathfrak{b}}(\ell_1^2(\mathbb{R})) = \limsup_{k\to\infty}(\mathbf{c}_{\mathfrak{b}}(k, \ell_1^2(\mathbb{R})))^{\frac{1}{k}},$$

the **complexification constant**. For a more thorough introduction to complexification constant and their basic properties, we refer the reader to [11].

Although we do not know the exact value of $\mathbf{c}_{\mathfrak{b}}(\ell_1^2(\mathbb{R}))$, in [3, Proposition 2.6] the following estimation was obtained

$$\sqrt[4]{2} \leq \mathbf{c}_{\mathfrak{b}}(\ell_1^2(\mathbb{R})) \leq \sqrt{2}.$$

Using the complexification constants as a tool, we can state and prove our main result.

Figure 1: Definition, exact-value question, and published bounds in arXiv:2311.03531v2, PDF page 3.

3 Proof intuition

The upper bound is a one-complex-variable argument. Restrict a real homogeneous polynomial to the complex line $x + \zeta y$ and apply Poisson–Jensen at $\zeta = i$. Maximizing the resulting logarithmic integral over the complex ℓ_1 unit sphere is an elementary concavity problem; the unique geometric maximum occurs when the two coordinates have equal modulus and phase difference $\pi/2$.

For sharpness, rotate the real diamond into a square by $u = x + y$, $v = x - y$. Homogeneous polynomials then become one-variable polynomials bounded by $\max(1, |t|)^k$ on the real line. A Cayley transform turns this into a compact weighted problem on $\mathbb{T}$ with weight $W(w) = \max(|1 - w|, |1 + w|)$. Its weighted extremal at the disk center is the Poisson mean of $\log W$, namely $2G/\pi$. The only subtle point is to show that this Poisson solution is the global weighted extremal; an explicit positive jump-density computation supplies that check.

4 The sharp upper bound

Let P be a real k-homogeneous polynomial on $\ell_1^2(\mathbb{R})$ with $\|P\| \leq 1$, and take $z = x + iy \in \ell_1^2(\mathbb{C})$ with $\|z\|_1 \leq 1$. The latter is precisely the Bochnak complexification used in the source [1, p. 3]. Apply Poisson–Jensen in the upper half-plane to

$$F(\zeta) = \widetilde{P}(x + \zeta y).$$

For real t, homogeneity and the real norm bound give $|F(t)| \leq \|x + ty\|_1^k$. Therefore

$$\log |\widetilde{P}(z)| \leq \frac{k}{\pi} \int_{\mathbb{R}} \frac{\log \|x + ty\|_1}{1 + t^2}\, dt. \tag{1}$$

Zeros only improve the inequality; degenerate cases follow by continuity.

Multiplication of z by a common unimodular scalar preserves its norm and changes $\widetilde{P}(z)$ only by a unimodular factor. On the unit sphere we may consequently write

$$z = (r, se^{i\delta}), \qquad r, s \geq 0, \qquad r + s = 1.$$

With $t = \tan\theta$ and $g(\theta) = |\cos\theta|$, the integral in (1), divided by k, becomes

$$J(r,\delta) = \log 2 + \frac{1}{\pi}\int_{-\pi/2}^{\pi/2} \log\bigl(rg(\theta) + sg(\theta - \delta)\bigr)\, d\theta. \tag{2}$$

For fixed δ, the integral is concave in r. Reflection $\theta \mapsto \delta - \theta$ shows that it is unchanged by $r \leftrightarrow 1 - r$, hence it is maximized at $r = s = 1/2$.

The evenness and π-periodicity of g reduce the phase difference to $\delta = 2h$ with $0 \leq h \leq \pi/4$. Splitting the integral at the zeros of the cosine terms yields

$$\begin{aligned} J(1/2, 2h) &= \log 2 + A(h), \\ A(h) &= \frac{2}{\pi}\Biggl[\Bigl(\frac{\pi}{2} - h\Bigr)\log\cos h + h\log\sin h \\ &\qquad\qquad + \int_0^{\pi/2-h} \log\cos t\, dt + \int_0^h \log\cos t\, dt\Biggr]. \end{aligned}$$

Differentiation cancels all four free logarithmic terms and gives

$$A'(h) = \frac{2}{\pi}\left[h\cot h - \Bigl(\frac{\pi}{2} - h\Bigr)\tan h\right].$$

This is nonnegative exactly when

$$h \geq \frac{\pi}{2}\sin^2 h. \tag{3}$$

On $[0, \pi/4]$ the function $\sin^2 h$ is convex, so it lies below the chord joining $(0,0)$ and $(\pi/4, 1/2)$; this is precisely (3). Thus the maximum occurs at $h = \pi/4$. The standard Catalan integral

$$\int_0^{\pi/4} \log\cos t\, dt = -\frac{\pi}{4}\log 2 + \frac{G}{2}$$

then gives

$$\max_{r,\delta} J(r,\delta) = \frac{2G}{\pi} - \frac{1}{2}\log 2. \tag{4}$$

Combining (1) and (4) proves, for every k,

$$c_{\mathrm{b}}(k, \ell_1^2(\mathbb{R})) \leq \left(\frac{e^{2G/\pi}}{\sqrt{2}}\right)^k. \tag{5}$$

5 A weighted polynomial lemma

Lemma 1. *For $k \geq 1$, let*

$$M_k = \sup\Bigl\{|p(i)| : \deg p \leq k, \quad |p(t)| \leq \max(1, |t|)^k \quad (t \in \mathbb{R})\Bigr\},$$

where complex coefficients are allowed. Then

$$\lim_{k\to\infty} M_k^{1/k} = e^{2G/\pi}.$$

The same limit holds when only real-coefficient polynomials are allowed.

Proof. Use the Cayley transform

$$w = \frac{t-i}{t+i}, \qquad t = i\frac{1+w}{1-w},$$

and set

$$q(w) = (1-w)^k p\left(i\frac{1+w}{1-w}\right).$$

Then $\deg q \le k$, $q(0) = p(i)$, and the real-line constraint is equivalent to

$$|q(w)| \le W(w)^k \quad (w \in \mathbb{T}), \qquad W(w) = \max(|1-w|, |1+w|). \tag{6}$$

Put $\varphi(\theta) = \log W(e^{i\theta})$ and let H be its Poisson extension to the unit disk. Since

$$W(e^{i\theta}) = 2\max\bigl(|\sin(\theta/2)|, |\cos(\theta/2)|\bigr),$$

symmetry and the Catalan integral give

$$H(0) = \frac{1}{2\pi}\int_0^{2\pi} \varphi(\theta)\, d\theta = \frac{4}{\pi}\int_0^{\pi/4} \log(2\cos u)\, du = \frac{2G}{\pi}. \tag{7}$$

We next verify that H is not merely a harmonic upper bound but the weighted extremal. Define on the plane

$$U(w) = \begin{cases} H(w), & |w| \le 1, \\ \log|w| + H(1/\overline{w}), & |w| \ge 1. \end{cases}$$

It is continuous, has logarithmic growth, and equals φ on $\mathbb{T}$. Its distributional Laplacian off $\mathbb{T}$ vanishes. Across $\mathbb{T}$, the outward radial derivative jump is

$$1 - 2\Lambda\varphi(\theta),$$

where Λ is the circular Dirichlet-to-Neumann operator. On $-\pi/2 < \theta < \pi/2$ one has $\varphi(\theta) = \log(2\cos(\theta/2))$, and a direct periodic Hilbert-transform calculation gives

$$1 - 2\Lambda\varphi(\theta) = \frac{2}{\pi}\frac{\operatorname{artanh}(\sin\theta)}{\sin\theta} \ge 0. \tag{8}$$

The quotient has limiting value 1 at zero and the expression extends π-periodically; its logarithmic endpoint singularities are integrable. For reference, if $b = \tan(\theta/2)$, the principal-value integral reduces to

$$\Lambda\varphi(\theta) = \frac{1}{2} - \frac{1}{\pi}\operatorname{artanh}(b)(b + b^{-1}) = \frac{1}{2} - \frac{1}{\pi}\frac{\operatorname{artanh}(\sin\theta)}{\sin\theta}.$$

Thus (8) proves that U is subharmonic on $\mathbb{C}$ and belongs to the Lelong class.

Let $V_{\mathbb{T},\varphi}$ be the weighted Siciak extremal. The function U is admissible, so $V_{\mathbb{T},\varphi} \ge H$ in the disk. Conversely, every Lelong-class subharmonic function bounded above by φ on $\mathbb{T}$ is at most the Poisson solution H in the disk. Hence

$$V_{\mathbb{T},\varphi} = H \qquad \text{on the unit disk.} \tag{9}$$

The weighted Siciak–Zakharyuta theorem for a compact set and a continuous weight identifies this extremal with the logarithm of the asymptotic weighted polynomial extremal [3, Theorem 1.1, standard grading]. At zero, (6), (7), and (9) therefore give

$$\lim_{k\to\infty} M_k^{1/k} = e^{H(0)} = e^{2G/\pi}.$$

Finally write any admissible complex-coefficient polynomial as $p = a + ib$ coefficientwise, where a, b have real coefficients. On $\mathbb{R}$ both $|a|$ and $|b|$ are bounded by $|p|$, and

$$\max(|a(i)|, |b(i)|) \geq \frac{1}{2}|p(i)|.$$

The constant factor disappears after taking kth roots. This proves the same asymptotic statement for real coefficients. $\square$

6 Sharpness and conclusion

Use real linear coordinates

$$u = x + y, \qquad v = x - y.$$

Then

$$|x| + |y| = \max(|u|, |v|). \tag{10}$$

Given a real polynomial p_k supplied by Lemma 1, form the real k-homogeneous polynomial

$$Q_k(u, v) = u^k p_k(v/u),$$

interpreted by its homogeneous coefficient expansion. For real u, v, the defining bound for p_k gives

$$|Q_k(u, v)| \leq \max(|u|, |v|)^k.$$

Thus $P_k(x, y) = Q_k(x + y, x - y)$ has norm at most one on the real ℓ_1 unit ball by (10).

Evaluate its complexification at

$$z_0 = (1/2, i/2), \qquad \|z_0\|_1 = 1.$$

At this point

$$u_0 = \frac{1+i}{2}, \qquad v_0 = \frac{1-i}{2}, \qquad \frac{v_0}{u_0} = -i, \qquad |u_0| = 2^{-1/2}.$$

Real coefficients imply $|p_k(-i)| = |p_k(i)|$, so

$$\|\widetilde{P}_k\| \geq 2^{-k/2}|p_k(-i)|.$$

Lemma 1 yields

$$\limsup_{k\to\infty} c_{\mathrm{b}}(k, \ell_1^2(\mathbb{R}))^{1/k} \geq \frac{e^{2G/\pi}}{\sqrt{2}}.$$

Together with (5), this proves Theorem 1.

7 Verification and novelty audit

1. **Poisson–Jensen sign and normalization.** For $F(\zeta) = \zeta + i$, the formula at i is an equality with value $\log 2$; this checks the half-plane kernel and the absence of a missing growth term.

2. **Geometric maximization.** Concavity first eliminates the coordinate moduli, and the derivative computation then eliminates the phase. The maximizing point is exactly $(1/2, i/2)$, the point used for the lower bound.

3. **Weighted-extremal gluing.** A first proof draft used the unbounded real-line weight directly. The compact circle reformulation and the positive density (8) remove both the weak-admissibility issue at infinity and the possible failure of harmonic gluing.

4. **Coefficient field.** Passing from complex to real coefficients costs at most a factor of two, which is harmless for the asymptotic constant.

5. **Numerical check.** Independent semi-infinite LP reconnaissance at z_0 gives kth roots 1.19627, 1.22386, 1.23090, and 1.23595 for $k = 12, 20, 24, 28$, respectively, increasing toward $1.2668686\ldots$. These computations are not used in the proof.

A bounded novelty search on August 17, 2026 used the arXiv identifier, the exact problem language, the decimal value, the Catalan-constant formula, and the terms "Bochnak complexification" and "ℓ_1^2". The 2025 follow-up paper [2] proves that infinite-dimensional ℓ_1 has the largest Bochnak constants, but its Proposition 2.8 still records only the $2^{1/4}$ lower bound for ℓ_1^2 and contains no Catalan-constant formula. No prior resolution of this exact-value problem was found. The main novelty risk is therefore an uncatalogued equivalent computation in weighted approximation theory.

References

[1] J. T. Rodríguez, *A 2-dimensional real Banach space with constant of analyticity less than one*, arXiv:2311.03531v2 (2025).

[2] J. T. Rodríguez, *On the norm of the complexification of polynomials*, Studia Math. **282** (2025), 1–23, doi:10.4064/sm230615-15-4.

[3] B. S. Magnússon, À. E. Sigurðardóttir, and R. Sigurðsson, *Polynomials with exponents in compact convex sets and associated weighted extremal functions: the Siciak–Zakharyuta theorem*, arXiv:2305.08260v3 (2024).